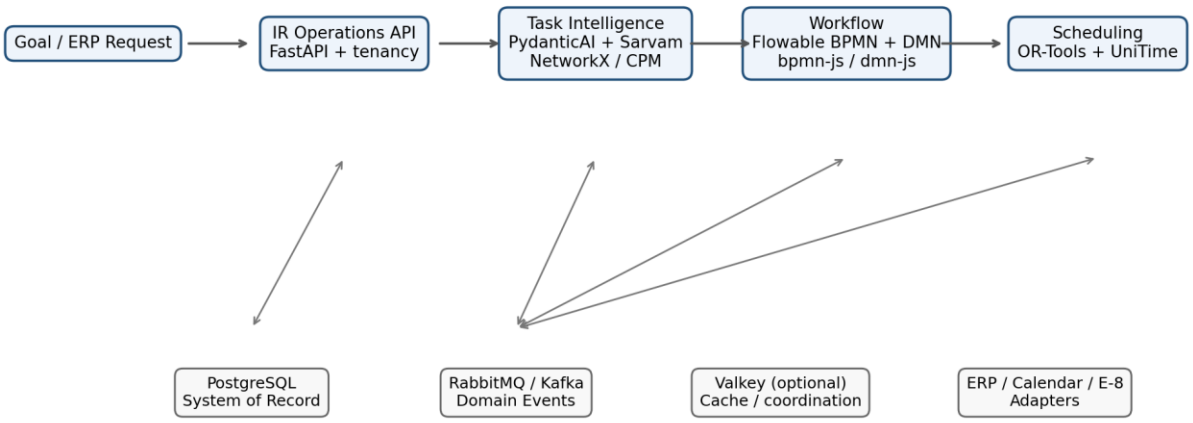
IR INFOTECH

E-7 OPERATIONS ENGINE

Open-Source Project Research, Selection & Adoption Report

Source-code links, licensing review, architecture fit, implementation mapping and final stack recommendation

Research date: 15 September 2026
Target: E-7 Operations Engine = Task Intelligence + Workflow + Scheduling
Status: Technology selection / PoC decision report



Recommended OSS composition: IR owns the orchestration/intelligence layer; proven OSS owns deterministic runtimes.

Prepared using the IR source documents as the functional source of truth and current official project repositories/documentation for external verification.

Document Control and Research Basis

Primary source of truth: IR_Infotech_E7_Operations_Engine_Complete_Specification(2).pdf

Underlying IR documents: IR MAIN DOC(10); Consolidated Engine Directory(5); Engine Architecture - IR Infotech AI Ecosystem(4); Task Intelligence story; Workflow story; Scheduling story; E-7 Comprehensive Implementation Plan

External research scope: Official repositories, official licensing files, release pages, official documentation and model repositories available as of 15 September 2026

Method boundary: The IR documents define what E-7 must do. External web research is used only to determine which existing projects can satisfy those requirements. Where a project does not publish a fixed hardware minimum, this report does not invent one; it calls for PoC benchmarking instead.

Source-of-Truth Requirements

- E-7 is an intelligence and orchestration layer, not a replacement ERP/project-management suite.
- The engine must implement the complete Plan → Automate → Schedule → Monitor loop.
- LLM reasoning may interpret goals and propose plans, but deterministic engines remain authoritative for workflow state, approval rules, SLA clocks and scheduling constraints.
- The solution must be API-first, platform-agnostic, tenant-aware, auditable and suitable for Embedded and Universal modes.
- IR requires market-first research and adoption/extension of mature open-source software before deciding to build from scratch.
- The final system must support WBS planning, dependencies/CPM, effort/resource recommendations, approvals, BPMN-style workflow automation, rules, SLA/escalation, conflict-free scheduling, incremental repair, calendar integration and event-driven operation.

IR Mandatory Evaluation Dimensions Used in This Report

#	Dimension	How it is applied here
1	Tool	Exact project and current release/branch when verified.
2	Description	Architecture and primary role.
3	Licensing	OSS/open-weight/source-available classification and product risk.
4	Tech stack	Language/runtime and integration model.
5	Resource utilization	Known runtime profile; PoC benchmark where no fixed official minimum exists.
6	Pros & Cons	Operational strengths and limitations for E-7.
7	Features & Applications	Direct mapping to E-7 functional requirements.
8	Comparison & Summary	Primary / conditional / reference / rejected decision.
9	References / Links	Official source code, model files, licensing and docs.

1. Executive Research Conclusion

No single open-source project should become the E-7 Operations Engine. The best architecture is a composed system in which IR Infotech owns the canonical intelligence/orchestration layer and mature OSS projects own specialized deterministic runtimes. This directly follows the IR requirement to adopt proven technology rather than rebuild commodity mechanisms.

Final selection: Recommended primary composition: PydanticAI + Sarvam-105B benchmark + NetworkX/CPM for Task Intelligence; Flowable BPMN/DMN with bpmn-js/dmn-js/form-js for Workflow; OR-Tools CP-SAT plus a UniTime PoC for Scheduling; PostgreSQL as system of record; RabbitMQ initially for events; Valkey only if distributed cache/coordination becomes necessary.

1.1 Selection Summary

E-7 area	Project	Decision	License	Why
Task Intelligence	PydanticAI + IR planner	**PRIMARY**	MIT	Typed structured generation, tools, validation-friendly agent framework.
Dependency / CPM	NetworkX + IR CPM logic	**ADOPT**	BSD-3-Clause	Deterministic DAG validation, topological ordering and graph calculations.
Workflow runtime	Flowable Engine	**PRIMARY**	Apache-2.0	BPMN/CMMN/DMN, human/system tasks, timers, REST/Java APIs, persistence.
Workflow UI	bpmn-js + dmn-js + form-js	**ADOPT**	bpmn.io License	Embeddable visual workflow, decision-table and form editors; watermark condition.
Generic scheduling	Google OR-Tools CP-SAT	**PRIMARY**	Apache-2.0	Strong deterministic

E-7 area	Project	Decision	License	Why
				constraint optimization; Python-friendly.
Academic scheduling	UniTime	**PRIMARY POC**	Apache-2.0	Course/exam/event/student scheduling, existing-timetable repair and SIS integration.
Durable code workflows	Temporal	**CONDITIONAL**	MIT	Add only if Flowable service tasks are insufficient for long-running code orchestration.
Java/Kotlin optimization	Timefold Solver	**CONDITIONAL**	Apache-2.0	Excellent domain-model constraint solver; alternative to OR-Tools where it simplifies modeling.
Advanced AI graph orchestration	LangGraph	**CONDITIONAL**	MIT	Introduce when Task Intelligence evolves into a truly stateful multi-agent planner.
School timetable fallback	FET 7.10.5	**BENCHMARK**	AGPL-3.0	Very rich educational timetable constraints; integration and licensing require care.
Task/project reference	OpenProject	**REFERENCE**	GPL-3.0	Excellent work-package/dependency/Gantt semantics and optional integration target.
Resource planning reference	LibrePlan	**REFERENCE**	AGPL-3.0	WBS, skills/resources, overload and project-planning concepts.
Modern task UI reference	Plane	**REFERENCE**	AGPL-3.0	Modern open-source project/task UI and API ecosystem.
ERP integration reference	Odoo / ERPNext	**REFERENCE**	LGPL/GPL	Useful adapters and ERP semantics, not E-7 core runtime.
Workflow fallback	Operaton	**CONDITIONAL**	Apache-2.0	Strong BPMN fallback if Flowable proves unsuitable.
Workflow candidate	Camunda 8	**REJECT AS FOUNDATION**	Mixed / Camunda License	Core production components are not uniformly permissive OSS.
Automation candidate	n8n	**REJECT AS FOUNDATION**	Sustainable Use / Enterprise	Source-available with commercial-use restrictions; poor fit for embedded IR product.

1.2 Architecture Decision

- Do not fork a full project-management product and call it E-7. That would violate the “intelligence layer” boundary and create unnecessary UI/domain coupling.
- Do not make an LLM authoritative for workflow transitions, SLAs or schedule generation. Structured AI suggestions are validated before persistence.

- Do reuse workflow and solver runtimes aggressively. These are mature engineering problems with high reliability cost if rebuilt.
- Keep a canonical E-7 domain model so Flowable, UniTime, OR-Tools, ERP systems and calendars can be changed behind adapters without changing the public API.

2. Primary Projects Recommended for Adoption

2.1 Flowable Engine

Decision: **PRIMARY** | E-7 role: Primary workflow/process runtime

Observed version / release: 8.0.0 (latest release observed)

License: Apache License 2.0

Technology stack: Java; BPMN, CMMN, DMN; Spring integration; Java and REST APIs

Source code: https://github.com/flowable/flowable-engine

Official documentation: https://www.flowable.com/open-source-code

E-7 fit

Covers BPMN process execution, human/system activities, gateways, timers, persisted process state, DMN decisions and REST integration. It maps directly to E-7 approval chains, deterministic state transitions, SLA timers, audit/history and workflow API requirements. [W01][W02]

Resource profile

JVM service. Exact production memory/CPU depends on concurrency, database load and history retention; benchmark under the expected number of process instances and timers.

Strengths

- Permissive Apache-2.0 core
- Mature BPMN/CMMN/DMN engine family
- REST and Java API options
- Good separation between workflow definition and runtime state
- Strong fit for human approvals and long-running institutional processes

Limitations / risks

- Adds a Java service to a Python-heavy stack
- Flowable OSS 7+ focuses primarily on engines/APIs rather than a complete free visual product suite
- Requires adapter code to map Flowable task/process IDs to canonical E-7 entities

Recommended integration pattern

Run as a dedicated Java service. E-7 keeps canonical workflow definitions/instances in PostgreSQL and stores Flowable deployment/instance references. Use BPMN/DMN artifacts generated by the IR UI. Service tasks call E-7 endpoints using idempotency keys.

2.2 bpmn-js / dmn-js / form-js

Decision: **ADOPT** | E-7 role: Visual workflow, decision and form authoring

Observed version / release: bpmn-js 18.22.1 observed; dmn-js/form-js current main releases

License: bpmn.io License (permissive-style with mandatory visible bpmn.io watermark)

Technology stack: JavaScript/TypeScript browser toolkits

Source code: https://github.com/bpmn-io/bpmn-js

dmn-js source: https://github.com/bpmn-io/dmn-js

form-js source: https://github.com/bpmn-io/form-js

Official documentation: https://bpmn.io/

E-7 fit

Provides embeddable editors so IR can build its own workflow designer, decision-table editor and task-form authoring UI while Flowable remains the execution runtime. [W03][W04][W05][W06]

Resource profile

Client-side browser workload; resource usage depends mainly on diagram size and active editor plugins. No separate server is required for the editor itself.

Strengths

- Purpose-built BPMN/DMN editing
- Embeddable and highly extensible
- Avoids dependence on a proprietary workflow-modeler frontend
- Natural match to Flowable BPMN/DMN assets

Limitations / risks

- The bpmn.io watermark must remain visible
- Requires IR frontend engineering for permissions, templates, deployment actions and versioning UX
- Not an execution engine

Recommended integration pattern

Embed in IR Operations UI. Save BPMN/DMN/form source as versioned artifacts, validate server-side, then deploy approved versions to Flowable.

2.3 Google OR-Tools CP-SAT

Decision: **PRIMARY** | E-7 role: Primary generic scheduling and constraint optimization solver

Observed version / release: 9.15

License: Apache License 2.0

Technology stack: C++ core with Python, Java, C# interfaces; CP-SAT and other optimization algorithms

Source code: https://github.com/google/or-tools

Official documentation: https://developers.google.com/optimization/

E-7 fit

Directly implements the deterministic scheduling boundary defined by E-7. It can model hard/soft constraints, objective penalties, resource assignments, precedence, no-overlap, capacities and schedule repair objectives. [W07][W08]

Resource profile

CPU and RAM scale with model size, constraints, horizon and solve-time budget. GPU is not required. Use dedicated solver workers and enforce per-job time budgets.

Strengths

- Permissive Apache-2.0
- Excellent Python integration
- Strong CP-SAT performance and mature examples
- General-purpose across education, HR and enterprise scheduling
- Allows explicit infeasibility/solution status and reproducible solver configuration

Limitations / risks

- IR must translate domain rules into a correct constraint model
- Constraint explainability must be added at the E-7 layer
- Large optimization models need careful profiling and decomposition

Recommended integration pattern

The Scheduling Service builds CP-SAT models from canonical resources, availability, assignments and constraint records. Solver workers return assignments, objective components and diagnostics; publishing remains a separate human-controlled action.

2.4 UniTime

Decision: PRIMARY | E-7 role: Primary education-specific scheduling project to PoC

Observed version / release: 4.9 build 135 (latest release observed)

License: Apache License 2.0

Technology stack: Java web application / scheduling suite with database-backed educational domain models

Source code: https://github.com/UniTime/unitime

Official documentation: https://help.unitime.org/

E-7 fit

UniTime covers course timetabling, examination timetabling, event management, student scheduling and timetable changes. This is significantly closer to E-7 educational scheduling than a generic project-management application and should be tested before IR writes a complete school/university timetable model from scratch. [W09][W10]

Resource profile

JVM/web/database deployment. Hardware depends strongly on institution size and solver use; benchmark with representative course/exam datasets.

Strengths

- Apache-2.0
- Purpose-built educational scheduling
- Course and exam scheduling in one platform
- Existing schedule change/repair concepts
- External/SIS integration patterns and rich domain semantics

Limitations / risks

- Education-specific; cannot replace generic OR-Tools scheduling
- Large established application with its own domain model, so adapter boundaries matter
- May be more application than library for some E-7 use cases

Recommended integration pattern

Treat UniTime as an education scheduling service behind an adapter. E-7 owns canonical resources/events and external IDs. Compare UniTime vs an OR-Tools-native school model on the same benchmark dataset before selecting the production education path.

2.5 PydanticAI

Decision: PRIMARY | E-7 role: Primary Task Intelligence agent framework for MVP

Observed version / release: 2.43.0 (11 Sep 2026)

License: MIT

Technology stack: Python, Pydantic-based typed agents/tools/models; integrates with multiple model providers

Source code: https://github.com/pydantic/pydantic-ai

Official documentation: https://ai.pydantic.dev/

E-7 fit

E-7 requires strict structured WBS output, typed validation, bounded repair and human review. PydanticAI fits that pattern with less orchestration overhead than a multi-agent graph framework for the first production slice. [W11][W12]

Resource profile

Framework overhead is small relative to model inference. Production resource use is dominated by the selected LLM/provider and retrieval/tool calls.

Strengths

- MIT
- Python-native
- Strong typed-output alignment with Pydantic schemas
- Provider flexibility
- Good fit for tool calls and validation-driven repair

Limitations / risks

- Does not provide E-7 planning intelligence by itself; prompts/schemas/evaluation remain IR IP
- Complex long-running multi-agent plans may eventually need a graph/persistence layer
- Model quality remains the main determinant of WBS quality

Recommended integration pattern

Goal intake → retrieve approved examples/context → PydanticAI planner emits PlanSchema → Pydantic/JSON Schema validation → NetworkX dependency validation → targeted repair → human approval.

2.6 NetworkX

Decision: **ADOPT** | E-7 role: Dependency graph, cycle detection and graph analytics

Observed version / release: 3.6.1

License: BSD 3-Clause

Technology stack: Python graph algorithms

Source code: https://github.com/networkx/networkx

Official documentation: https://networkx.org/

E-7 fit

Ideal for DAG validation, topological ordering, dependency traversals and graph calculations around WBS plans. Critical Path Method calculations can be implemented on top of the validated DAG with auditable deterministic logic. [W13][W14]

Resource profile

In-memory Python graph structures; resource use scales with nodes/edges. E-7 task graphs are expected to be much smaller than the large-network cases NetworkX targets, but benchmark unusually large portfolios.

Strengths

- Permissive BSD
- Python-native
- Mature graph algorithms
- Simple integration with FastAPI/Pydantic models
- Useful for cycle paths, successors/predecessors and analytics

Limitations / risks

- CPM/business scheduling semantics remain IR code

- Not a persistence layer
- Very large graphs may need specialized alternatives, unlikely for typical single-operation WBS

Recommended integration pattern

Build a dependency service that materializes the current plan graph, rejects cycles, computes topological order and exposes critical-path/slack calculations with deterministic unit tests.

3. Core Open-Source Infrastructure

Project	Decision	License	E-7 use	Source code
PostgreSQL	PRIMARY	PostgreSQL License	Transactional system of record for canonical goals/plans/tasks/workflow mappings/resources/schedules/audit.	Repository
RabbitMQ	PRIMARY INITIAL BROKER	MPL 2.0	Initial event broker for task/workflow/SLA/schedule events and asynchronous jobs.	Repository
Apache Kafka	CONDITIONAL	Apache-2.0	High-throughput event-streaming alternative if the broader IR ecosystem standardizes on Kafka.	Repository
Valkey	OPTIONAL	BSD 3-Clause	Optional cache, ephemeral job status and coordination layer.	Repository

PostgreSQL — The public GitHub repository is an official mirror; PostgreSQL remains the authoritative transactional boundary for E-7. Source | Docs

RabbitMQ — Prefer for the first deployment unless IR has a platform-wide Kafka standard or very high event-stream requirements. Source | Docs

Apache Kafka — More operationally expensive than RabbitMQ for a small first slice; adopt when stream retention/replay/throughput requirements justify it. Source | Docs

Valkey — For an open-source-first stack, Valkey is a cleaner default than relying on vendor-specific Redis licensing. E-7 can often start with PostgreSQL advisory locks and add Valkey only after measured need. Source | Docs

4. Conditional / Secondary Projects

4.1 Temporal

Decision: CONDITIONAL | Durable code-first orchestration for long-running distributed activities.

Observed version / state: 1.32.0 observed

License: MIT

Source code / source archive: https://github.com/temporalio/temporal

Official docs: https://docs.temporal.io/

Temporal is excellent at durable execution and automatic recovery, but E-7 already needs a BPMN/human workflow engine. Starting with both Flowable and Temporal increases operational complexity. Add Temporal only when real workflows contain long-running code pipelines that are awkward as Flowable service tasks.

Integration decision: Flowable stays authoritative for business/human process state. A Flowable service task may start a Temporal workflow for a code-heavy durable sub-process and receive a completion callback/event.

4.2 Timefold Solver

Decision: CONDITIONAL | Java/Kotlin constraint solver alternative to OR-Tools.

Observed version / state: 2.3.0 observed in release list

License: Apache-2.0

Source code / source archive: https://github.com/TimefoldAI/timefold-solver

Official docs: https://solver.timefold.ai/

Strong choice for employee rostering, task assignment and other planning problems where a Java domain model and Constraint Streams make the constraints easier to maintain than CP-SAT. Do not run two general solvers in the MVP without evidence that both are needed.

Integration decision: Benchmark one representative allocation problem against OR-Tools. Select Timefold only if developer productivity, score explainability or Java ecosystem integration materially improves.

4.3 LangGraph

Decision: CONDITIONAL | Stateful graph orchestration for advanced Task Intelligence.

Observed version / state: langgraph 1.2.11 appears in current release list

License: MIT

Source code / source archive: https://github.com/langchain-ai/langgraph

Official docs: https://docs.langchain.com/oss/python/langgraph/overview

Useful for durable, branching, human-in-the-loop AI agent graphs. For the first E-7 planner, a typed PydanticAI flow is simpler and easier to validate. Introduce LangGraph when planning becomes a multi-agent/critic/reviewer graph with persisted checkpoints.

Integration decision: Potential future graph: goal analyzer → policy retriever → planner → dependency critic → resource critic → risk critic → repair → human approval.

4.4 FET

Decision: BENCHMARK | Specialized school/university timetable generator.

Observed version / state: 7.10.5 (11 Sep 2026)

License: AGPL-3.0

Source code / source archive: https://lalescu.ro/liviu/fet/download.html

Official docs: https://lalescu.ro/liviu/fet/

FET has a very rich educational constraint catalog and is valuable as a benchmark/fallback. The current official source is distributed as a source archive on the FET site; public GitHub mirrors can be stale. AGPL and service-wrapper implications require legal review before embedding/distribution.

Integration decision: Run the same educational benchmark through FET, UniTime and an OR-Tools-native model. Use FET only if its constraint coverage and solve quality outweigh integration/licensing costs.

4.5 Operaton

Decision: CONDITIONAL | BPMN workflow fallback / alternative runtime.

Observed version / state: Current community project

License: Apache-2.0

Source code / source archive: https://github.com/operaton/operaton

Official docs: https://operaton.org/

Community-driven continuation of Camunda 7 concepts and a strong permissive alternative. Keep it as a fallback ADR if Flowable integration or roadmap becomes unsuitable.

Integration decision: Do not build dual runtime support initially; keep E-7 workflow adapter interfaces generic enough that an Operaton adapter remains possible.

5. Reference and Integration Platforms - Study, Do Not Fork as E-7

Project	Decision	License	Useful coverage	Source code	Research conclusion
OpenProject	REFERENCE / OPTIONAL ADAPTER	GPL-3.0	Work packages, hierarchy, dependencies, Gantt/project scheduling, progress UX and API integration.	Repository Docs	Use as task/project semantics reference and optional integration target; do not make it E-7 system of record.
LibrePlan	REFERENCE	AGPL-3.0	WBS planning, workers/machines, skills/roles, calendars and multi-project resource planning.	Repository Docs	Very useful research source for resource planning; AGPL and older architecture make it a poor E-7 foundation.
Plane	REFERENCE	AGPL-3.0	Modern task/project interface, self-hosting and API ecosystem.	Repository Docs	Use for modern operations/task UX ideas or optional integration, not BPMN or scheduling core.
Odoo Community	REFERENCE / ERP ADAPTER	Predominantly LGPL-3+ community code; module-specific licenses must be checked	ERP/project/resource reference and realistic external system for Embedded-mode adapters.	Repository Docs	Do not couple E-7 to Odoo domain tables; map Odoo entities through canonical adapters.
ERPNext	REFERENCE / ERP ADAPTER	GPL-3.0	Open-source ERP with project, task and workflow concepts.	Repository Docs	Useful integration target and reference for institutional workflows, but lacks E-7 AI planning and solver-based scheduling.

6. Candidates Rejected as the E-7 Foundation

Camunda 8

Decision: REJECT

Camunda 8 is technologically capable, but key core components are under the Camunda License rather than uniformly permissive OSS. This creates an avoidable production licensing dependency for IR's open-source-first engine strategy.

Source repository: https://github.com/camunda/camunda

n8n

Decision: REJECT

n8n is source-available under the Sustainable Use License, which restricts use to internal business/non-commercial scenarios unless licensed. It is excellent automation software but a poor foundation for an embedded commercial E-7 product.

Source repository: https://github.com/n8n-io/n8n

Licensing: This is a technical licensing screen, not legal advice. Any GPL/AGPL, bpmn.io watermark, model, or mixed-license component selected for distribution should receive legal review before a commercial release.

7. Task Intelligence Model Research - Sarvam-105B

Updated recommendation: benchmark Sarvam-105B rather than the earlier Sarvam-M reference. The current Sarvam-105B model repository is Apache-2.0 and publishes model files plus custom Transformers code on Hugging Face. The model card describes a 105B MoE model with 10.3B active parameters and 128K-context architecture. [W21]

Model repository / files: https://huggingface.co/sarvamai/sarvam-105b

License: Apache-2.0

Use in E-7: Candidate reasoning model for goal interpretation, WBS generation, assumptions and explanation; never authoritative for workflow/scheduling state.

Deployment note: Benchmark via hosted/API access first unless IR has hardware appropriate for a large 105B MoE model. Self-hosting must be a measured infrastructure decision, not assumed.

- Structured-output validity
- Required-task coverage
- Invalid dependency rate
- Hallucinated resource/entity rate
- Human correction rate
- Latency and cost
- Fallback behavior under timeout/invalid JSON

8. E-7 Requirement-to-Project Mapping

Requirement	Recommended project(s)	Implementation boundary
Goal → structured WBS	PydanticAI + Sarvam-105B + IR schemas	IR owns planning prompts, schema, examples, validation and evaluation.
Dependency cycle detection / topological order	NetworkX	Deterministic; model suggestions are not trusted until validated.
Critical Path / slack	NetworkX + IR CPM	Transparent calculations stored/versioned.
Effort estimation	IR heuristics → XGBoost/LightGBM later	Do not train ML until sufficient clean execution history exists.
BPMN process execution	Flowable	Primary workflow runtime.
Workflow visual designer	bpmn-js	Embedded in IR Operations UI.
Decision rules	Flowable DMN + dmn-js	Use before introducing Drools/Easy Rules.
Approval forms	form-js + IR form schema mapping	Human task forms bound to typed variables.
SLA timers / escalation	Flowable + IR SLA policy mapping	Persistent timers; events route to E-8.
Durable code-first subflows	Temporal (optional)	Only if demonstrated need.
Generic resource/time optimization	OR-Tools CP-SAT	Primary constraint solver.
Academic timetable / exams	UniTime PoC; OR-Tools alternative	Compare on same school dataset.

Requirement	Recommended project(s)	Implementation boundary
Educational timetable fallback	FET	Benchmark; AGPL/legal review.
Alternate solver	Timefold	Adopt only if Java domain modeling materially helps.
Task/project integration	OpenProject / Plane / ERP adapters	External systems remain external; canonical IDs in E-7.
System of record	PostgreSQL	Transactional canonical state and outbox.
Domain events	RabbitMQ initially / Kafka conditional	Versioned events and idempotent consumers.
Cache / coordination	Valkey optional	Add after measured need.

9. Licensing and Product-Risk Matrix

Project	License	IR risk	Guidance
Flowable	Apache-2.0	Low	Strong fit for embedded commercial use.
OR-Tools	Apache-2.0	Low	Permissive.
UniTime	Apache-2.0	Low	Permissive; inspect bundled third-party notices.
PydanticAI	MIT	Low	Permissive.
NetworkX	BSD-3-Clause	Low	Permissive.
Temporal	MIT	Low	Permissive.
Timefold Community	Apache-2.0	Low	Community edition permissive; enterprise edition separate.
LangGraph	MIT	Low	Open-source framework; commercial LangSmith services are separate.
bpmn.io toolkits	bpmn.io License	Medium	Permissive-style but watermark source must remain and watermark must stay visible.
RabbitMQ	MPL-2.0	Low/Medium	File-level copyleft; generally suitable as separate service.
PostgreSQL	PostgreSQL License	Low	Permissive.
Valkey	BSD-3-Clause	Low	Permissive.
OpenProject	GPL-3.0	Medium	Good external integration/reference; avoid copying code into proprietary core.
ERPNext	GPL-3.0	Medium	Treat as external system/integration target.
Odoo Community	LGPL-3+ / mixed modules	Medium	Check each module used.
LibrePlan	AGPL-3.0	High if embedded/modified	Prefer study/reference unless legal strategy supports AGPL.
Plane	AGPL-3.0	High if embedded/modified	Reference/integration rather than core.
FET	AGPL-3.0	High if embedded/distributed	Use service boundary/benchmark;

Project	License	IR risk	Guidance
			legal review.
Camunda 8 core	Camunda License / mixed	High	Rejected as open-source-first foundation.
n8n	Sustainable Use + Enterprise	High	Source available, not OSI open source; rejected as foundation.
Sarvam-105B	Apache-2.0 model repository	Low license / High infra	License friendly; deployment footprint must be benchmarked.

10. What IR Infotech Should Reuse vs Build

Capability	Reuse	IR-specific work
BPMN runtime	Flowable	Workflow adapter, canonical states, tenant policies, templates, deployment governance.
Workflow designer	bpmn-js/dmn-js/form-js	IR UX, permissions, versioning, template library, deployment controls.
Constraint solving	OR-Tools / UniTime / Timefold when selected	Canonical constraint catalog, model builder, explanation layer, repair policy.
Graph algorithms	NetworkX	E-7 dependency semantics, CPM/risk policy and persistence.
Agent framework	PydanticAI; LangGraph later	Goal decomposition prompts, PlanSchema, guardrails, repair loop, gold-set evaluation.
Project/task products	OpenProject/Plane/LibrePlan as references or integrations	Canonical task model, source mapping, cross-system synchronization.
Events	RabbitMQ/Kafka	Versioned domain-event schemas, transactional outbox, consumer idempotency.
Data store	PostgreSQL	E-7 schema, tenancy, migrations, audit model and adapter mappings.

Core IR intellectual property should be: the canonical platform-agnostic operations model; goal-to-plan intelligence; validation and confidence policy; cross-mode orchestration; ERP/calendar/engine adapters; constraint translation; tenant/security boundaries; operational event contracts; and explainability/audit correlation.

11. Final Recommended Stack

Layer	Selection	Decision
API / orchestration	FastAPI + Pydantic models	IR-owned canonical API and orchestration.
Task Intelligence	PydanticAI + Sarvam-105B benchmark + fallback model	Typed plan generation and repair.
Dependency / CPM	NetworkX + IR CPM	Deterministic graph checks.
Workflow runtime	Flowable Engine	BPMN/CMMN/DMN and process persistence.
Workflow UI	bpmn-js + dmn-js + form-js	IR visual authoring experience.
Generic scheduler	OR-Tools CP-SAT	Primary hard/soft constraint solver.
Education scheduling	UniTime PoC	Adopt if benchmark fit is strong; otherwise use OR-Tools-native model.

Layer	Selection	Decision
Durable code orchestration	Temporal only if required	Do not deploy on day one.
Database	PostgreSQL	Canonical source of truth.
Event broker	RabbitMQ first; Kafka if platform standard	Event-driven decoupling.
Cache	Valkey only after demonstrated need	Optional.
Observability	OpenTelemetry + Prometheus/Grafana or organization equivalents	Metrics/traces/log correlation.

11.1 Recommended PoC Sequence

1. PoC 1 - Task Intelligence: Build one goal → typed WBS → dependency validation → human approval path with PydanticAI, Sarvam-105B benchmark and NetworkX. Create a fixed gold set and compare against at least one fallback model.
2. PoC 2 - Workflow: Model one representative approval process in bpmn-js, deploy to Flowable, use DMN for at least one branching decision, test SLA warning/breach, restart recovery, authorization and audit.
3. PoC 3 - Scheduling: Use one shared school/exam dataset to compare OR-Tools, UniTime and FET. Measure feasibility, solve time, number of soft violations, repair stability and adapter effort.
4. PoC 4 - Vertical slice: Approved AI plan starts a Flowable process; completed tasks generate schedule demand; scheduler publishes assignments; a resource.unavailable event triggers targeted repair; every step remains auditable.
5. PoC 5 - Integration hardening: Connect one ERP/calendar system through canonical external IDs, idempotent APIs and transactional outbox; validate tenant isolation and recovery.

11.2 Go / No-Go Gates

- Task Intelligence: ≥ approved threshold for structured-output validity, required-task coverage and invalid-dependency rate on the gold set.
- Workflow: sequential/parallel approvals, SLA breach, retry/restart and duplicate callback tests all pass without duplicate side effects.
- Scheduling: zero hard-constraint violations on feasible regression datasets; explicit diagnostics on infeasible datasets; incremental repair preserves locked/unaffected assignments.
- Licensing: legal approval for every non-permissive or special-license component before product embedding/distribution.
- Operations: target APIs, jobs and event consumers expose logs, traces, metrics and correlation IDs.

12. Source Code and Official Project Link Directory

Project	Source code / model files	Official docs / homepage
Flowable Engine	https://github.com/flowable/flowable-engine	https://www.flowable.com/open-source-code
bpmn-js	https://github.com/bpmn-io/bpmn-js	https://bpmn.io/toolkit/bpmn-js/
dmn-js	https://github.com/bpmn-io/dmn-js	https://bpmn.io/toolkit/dmn-js/
form-js	https://github.com/bpmn-io/form-js	https://bpmn.io/toolkit/form-js/
Google OR-Tools	https://github.com/google/or-tools	https://developers.google.com/optimization/
UniTime	https://github.com/UniTime/unitime	https://help.unitime.org/
PydanticAI	https://github.com/pydantic/pydantic-ai	https://ai.pydantic.dev/

Project	Source code / model files	Official docs / homepage
NetworkX	https://github.com/networkx/networkx	https://networkx.org/
Temporal	https://github.com/temporalio/temporal	https://docs.temporal.io/
Timefold Solver	https://github.com/TimefoldAI/timefold-solver	https://solver.timefold.ai/
LangGraph	https://github.com/langchain-ai/langgraph	https://docs.langchain.com/oss/python/langgraph/overview
FET current official source archive	https://lalescu.ro/liviu/fet/download.html	https://lalescu.ro/liviu/fet/
Operaton	https://github.com/operaton/operaton	https://operaton.org/
OpenProject	https://github.com/opf/openproject	https://www.openproject.org/docs/
LibrePlan	https://github.com/LibrePlan/libreplan	https://www.libreplan.dev/
Plane	https://github.com/makeplane/plane	https://docs.plane.so/
Odoo Community	https://github.com/odoo/odoo	https://www.odoo.com/documentation/19.0/
ERPNext	https://github.com/frappe/erpnext	https://docs.frappe.io/erpnext
PostgreSQL GitHub mirror	https://github.com/postgres/postgres	https://www.postgresql.org/docs/
RabbitMQ	https://github.com/rabbitmq/rabbitmq-server	https://www.rabbitmq.com/docs
Apache Kafka	https://github.com/apache/kafka	https://kafka.apache.org/documentation/
Valkey	https://github.com/valkey-io/valkey	https://valkey.io/
Camunda 8 (rejected foundation)	https://github.com/camunda/camunda	https://docs.camunda.io/
n8n (rejected foundation)	https://github.com/n8n-io/n8n	https://docs.n8n.io/
Sarvam-105B model repository	https://huggingface.co/sarvamai/sarvam-105b	https://docs.sarvam.ai/

13. References

13.1 IR Source Documents

[S01] IR_Infotech_E7_Operations_Engine_Complete_Specification(2).pdf - complete E-7 product, architecture and implementation blueprint.

[S02] IR MAIN DOC(10).pdf - mandatory market-first research process, 9-dimension framework, platform principles and original engine requirements.

[S03] Consolidated Engine Directory(5).pdf - E-7 merge and Plan → Automate → Schedule logic.

[S04] Engine Architecture - IR Infotech AI Ecosystem(4).pdf - open-source-first philosophy, dual access modes and deterministic scheduling boundary.

[S05] 2task_intelligence(1).pdf - Task Intelligence features, deliverables and acceptance criteria.

[S06] 13workflow(1).pdf - Workflow features, deliverables and acceptance criteria.

[S07] 19scheduling(1).pdf - Scheduling features, deliverables and acceptance criteria.

[S08] E-7 Comprehensive Implementation Plan - recommended candidate technologies and reasoning/execution separation.

13.2 External Official Sources Used for Verification

[W01] Flowable source repository: https://github.com/flowable/flowable-engine

[W02] Flowable open-source overview: https://www.flowable.com/open-source-code

[W03] bpmn-js source: https://github.com/bpmn-io/bpmn-js

[W04] dmn-js source: https://github.com/bpmn-io/dmn-js

[W05] form-js source: https://github.com/bpmn-io/form-js

[W06] bpmn.io toolkit license: https://bpmn.io/license/

[W07] OR-Tools source: https://github.com/google/or-tools

[W08] OR-Tools releases: https://github.com/google/or-tools/releases

[W09] UniTime source: https://github.com/UniTime/unitime

[W10] UniTime releases: https://github.com/UniTime/unitime/releases

[W11] PydanticAI source: https://github.com/pydantic/pydantic-ai

[W12] PydanticAI releases: https://github.com/pydantic/pydantic-ai/releases

[W13] NetworkX source: https://github.com/networkx/networkx

[W14] NetworkX releases: https://github.com/networkx/networkx/releases

[W15] Temporal source: https://github.com/temporalio/temporal

[W16] Timefold source: https://github.com/TimefoldAI/timefold-solver

[W17] LangGraph source: https://github.com/langchain-ai/langgraph

[W18] FET official source download: https://lalescu.ro/liviu/fet/download.html

[W19] OpenProject source: https://github.com/opf/openproject

[W20] LibrePlan source: https://github.com/LibrePlan/libreplan

[W21] Sarvam-105B model repository: https://huggingface.co/sarvamai/sarvam-105b

[W22] Plane source: https://github.com/makeplane/plane

[W23] Odoo source: https://github.com/odoo/odoo

[W24] ERPNext source: https://github.com/frappe/erpnext

[W25] RabbitMQ source: https://github.com/rabbitmq/rabbitmq-server

[W26] Apache Kafka source: https://github.com/apache/kafka

[W27] Valkey source: https://github.com/valkey-io/valkey

[W28] PostgreSQL GitHub mirror: https://github.com/postgres/postgres

[W29] Camunda 8 source/license: https://github.com/camunda/camunda

[W30] n8n source/license: https://github.com/n8n-io/n8n

14. Final Decision Statement

Recommended implementation direction: build E-7 as a thin IR-owned operations intelligence layer over Flowable, OR-Tools/UniTime, PydanticAI/NetworkX and open infrastructure. Do not make a single third-party project the product, and do not rebuild mature BPM/solver mechanisms. The first production slice should intentionally use the minimum number of engines and add Temporal, Timefold, LangGraph, FET or a separate vector/cache service only after a measured requirement appears.

Actionable next step: The most important Phase-0 decisions are now concrete: Flowable vs Operaton fallback; OR-Tools vs UniTime for the first education scheduler; PydanticAI planner benchmark with Sarvam-105B and one fallback model; RabbitMQ vs existing platform Kafka; and legal acceptance of the bpmn.io watermark/license for the visual designer.